

Reliability auditing for structural routing of QAOA optimizers

Molena Huynh^{1,*}

¹*North Carolina State University, Raleigh, North Carolina 27695, USA*

(Dated: July 2026)

Learned routing among classical optimizers can underperform a fixed schedule. We audit depth-2 MaxCut QAOA as a target-free portfolio: nine arms share a 32-call ceiling, a structural ridge model predicts anytime regret, and an order-statistic rule abstains to a family control. After development on 64 graphs, one internally locked audit gave mean AURC 0.16930 versus 0.17207 on 160 new instances ($\Delta = -0.002768$). Twelve routes were accepted and four were harmful. Coverage was 141/160, below the frozen 0.90 gate. Under an iid-binomial sensitivity model this miss is compatible with nominal variation ($P = 0.249$), but the 95% upper risk bounds do not certify coverage-failure or conditional-harm control. Fixed quotas and sequential non-isomorphism filtering also preclude pooled exchangeability. The reliability gate therefore fails. This exact-simulator audit provides neither hardware nor scaling evidence.

Revision status. This manuscript supersedes the scientific claims and target-based evaluation of the first repository release and arXiv:2604.24803v1. The optimizer-portfolio evidence reported here uses new protocols, new seeds, a target-free metric and a separate internally locked confirmatory audit. The earlier target-hitting results are discussed only as motivation and are not combined with the new evidence.

Variational quantum algorithms combine a parameterized quantum circuit with a classical outer-loop optimizer. Each objective evaluation can require repeated circuit preparation and measurement, so optimization strategy multiplies the cost of an already expensive experimental loop [4, 5]. The quantum approximate optimization algorithm (QAOA) makes this interaction especially visible: parameters that work well on one graph can transfer to related graphs, while optimizer rankings can change with graph structure, circuit depth, noise and hyperparameters [1, 3, 14].

Parameter concentration, schedule interpolation, warm starts and graph neural networks exploit shared structure to initialize QAOA [6–10]. Learned optimizers go further by predicting trajectories or search policies [11, 12]. These approaches can reduce search effort, but they create a second statistical problem: deciding when a learned recommendation is safer than a strong, cheap control. Model selection on the same benchmark used to advertise performance is vulnerable to selection bias [20, 21]. Average gains also conceal selective harm; a router can improve a mean by helping one graph family while degrading another. Instance-level algorithm selection has already been posed for choosing between quantum and classical MaxCut solvers [22], while portfolio-learning theory shows that enlarging the portfolio can increase opportunity and overfitting simultaneously [23]. Our narrower problem routes among classical outer-loop strategies for one quantum objective and audits the resulting selective risk.

An earlier version of this project predicted a graph-conditioned Gaussian over QAOA angles and used it as an initializer, preconditioner, trust region and budget policy [13]. Its favorable comparison to random restart did not survive stronger controls: fixed concentration and simple schedule initializations were cheaper, and the stopping metric depended on an offline reference target unavailable in deployment. That negative result changes the question. Instead of asking whether one learned angle policy dominates, we ask whether graph structure can route among a deliberately heterogeneous *portfolio* while preserving a measurable reliability criterion.

We make four contributions. First, every arm receives an identical, target-free 32-query budget and is scored by an anytime normalized-regret curve rather than by time to a privileged target. Second, a low-capacity ridge model maps 16 relabeling-invariant graph summaries to per-arm utility; the strongest arm for each development family remains the deployable fallback. Third, a one-sided residual order statistic, motivated by the split-conformal construction, turns routing into selective prediction: the model abstains unless its estimated selected-minus-fallback regret plus that statistic is negative [24, 26]. Fourth, the graph protocol, implementation, non-audit traces and decision are content-hashed before audit evaluation, and an output lock rejects a second run in the same evidence directory.

The confirmatory conclusion is deliberately negative. A development study passed all frozen criteria, but the one-shot 160-graph audit did not. The prespecified 4,000-resample stratified bootstrap returned two negative percentile endpoints, but the upper endpoint is boundary-sensitive under reasonable alternative inference. Descriptive coverage also missed its threshold and four accepted routes were harmful. This distinction between the output of one locked resampling analysis and a failed reliability decision is the central result.

* molena.huynh@jmp.com

I. RESULTS

A. A target-free portfolio replaces privileged stopping

For an unweighted graph $G = (V, E)$, depth- p QAOA prepares

$$|\gamma, \beta\rangle = \prod_{j=p}^1 e^{-i\beta_j B} e^{-i\gamma_j C} |+\rangle^{\otimes |V|}, \quad (1)$$

where $C = \sum_{(u,v) \in E} (I - Z_u Z_v)/2$, $B = \sum_v X_v$, and the exact objective is $F_G(\theta) = \langle C \rangle_\theta$ [1, 2]. We use $p = 2$ and evaluate F_G exactly. An optimizer never receives a target value, exact MaxCut value or target-based stopping event. It receives a counter that permits exactly $B = 32$ objective calls.

At frozen checkpoints

$$\mathcal{T} = \{1, 2, 4, 8, 16, 32\}, \quad (2)$$

the runner records the best value observed so far. With exact MaxCut value $C_{\max}(G)$ used only after optimization, arm m has checkpoint regret

$$r_{im}(t) = 1 - \frac{\max_{s \leq t} F_{G_i}(\theta_{ims})}{C_{\max}(G_i)} \quad (3)$$

and primary utility

$$u_{im} = \text{AURC}_{im} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} r_{im}(t). \quad (4)$$

Lower AURC is better. Equal checkpoint weights emphasize useful values early without inventing a deployment target. Exact MaxCut is still a privileged small-instance analysis quantity; removing it from optimizer control does not make the metric hardware-native.

The portfolio contains nine fixed arms: a training-set concentration vector; four midpoint linear schedules with scale $d_t \in \{0.25, 0.50, 0.75, 1.00\}$; one-nearest-neighbor angle transfer; four-start random multistart; SPSA; and the legacy graph-conditioned Gaussian trust-region policy. The fixed schedules are included as separate arms because schedule choice can be competitive without instance-level learning [3, 16]. Every arm uses the full budget, and all seed evaluations are counted.

B. Structural routing abstains to a strong family fallback

For each graph we compute a 16-dimensional relabelling-invariant vector $x(G)$: size, edge count per vertex, density, seven degree summaries, mean clustering, transitivity, triangles per vertex, and three normalized Laplacian eigenvalue summaries. A multi-output ridge

regression fitted only on development graphs predicts every arm’s AURC,

$$\hat{u}(x) = \mathbf{a} + W\tilde{x}, \quad \lambda_{\text{ridge}} = 1, \quad (5)$$

where $\tilde{x}$ uses development means and scales. The proposed arm is $s(x) = \arg \min_m \hat{u}_m(x)$.

A family label is available for these synthetic generators. For family f , the fallback is the arm with lowest mean development AURC,

$$b(f) = \arg \min_m \frac{1}{N_f} \sum_{i:f_i=f} u_{im}. \quad (6)$$

This is intentionally stronger than falling back to a single global arm. In the heterogeneous study, Erdős–Rényi and Watts–Strogatz graphs fall back to the $d_t = 0.50$ schedule; random-regular and Barabási–Albert graphs fall back to $d_t = 0.75$.

On calibration graph i , define the selected-minus-fallback residual

$$R_i = [u_{i,s(x_i)} - u_{i,b(f_i)}] - [\hat{u}_{s(x_i)}(x_i) - \hat{u}_{b(f_i)}(x_i)]. \quad (7)$$

With $n_{\text{cal}} = 48$ and $\alpha = 0.10$, we use the one-sided split-conformal rank rule to choose the $\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil = 45$ th ordered residual, $q = 0.0358185$. For a new graph,

$$U(x) = \hat{u}_{s(x)}(x) - \hat{u}_{b(f)}(x) + q. \quad (8)$$

The deployed arm is $s(x)$ only when $s(x) \neq b(f)$ and $U(x) < 0$; otherwise it is $b(f)$. If calibration residuals and a future residual were exchangeable, the standard split-conformal theorem would provide marginal one-sided coverage $\mathbb{P}\{u_s - u_b \leq U\} \geq 1 - \alpha$, up to its finite-sample rank convention [24]. That premise is not exactly satisfied here: calibration and audit sets have fixed family-by-size quotas, and protocol construction sequentially rejects graphs isomorphic to any previously accepted graph across earlier splits. Thus q is a conformal-motivated order-statistic threshold, not a finite-sample coverage certificate for the realized audit; its empirical coverage is descriptive. Algebraically, the joint event “accepted and truly worse than fallback” is still contained in the observed event $u_s - u_b > U$, but no nominal probability bound follows for this protocol.

C. Development selected a heterogeneous protocol

We used development evidence only to decide whether a fresh confirmation was warranted. A first regular-grid study on 48 audit graphs improved the global baseline but not the development-selected family control: mean gated-minus-family AURC was $+0.000936$ with interval $[-0.002094, 0.004013]$. It accepted 13 graphs, harmed nine and achieved coverage 0.7708. This protocol failed and was not promoted.

A second development protocol varied generator parameters continuously or discretely within broad ranges.

An internally locked, target-free decision path

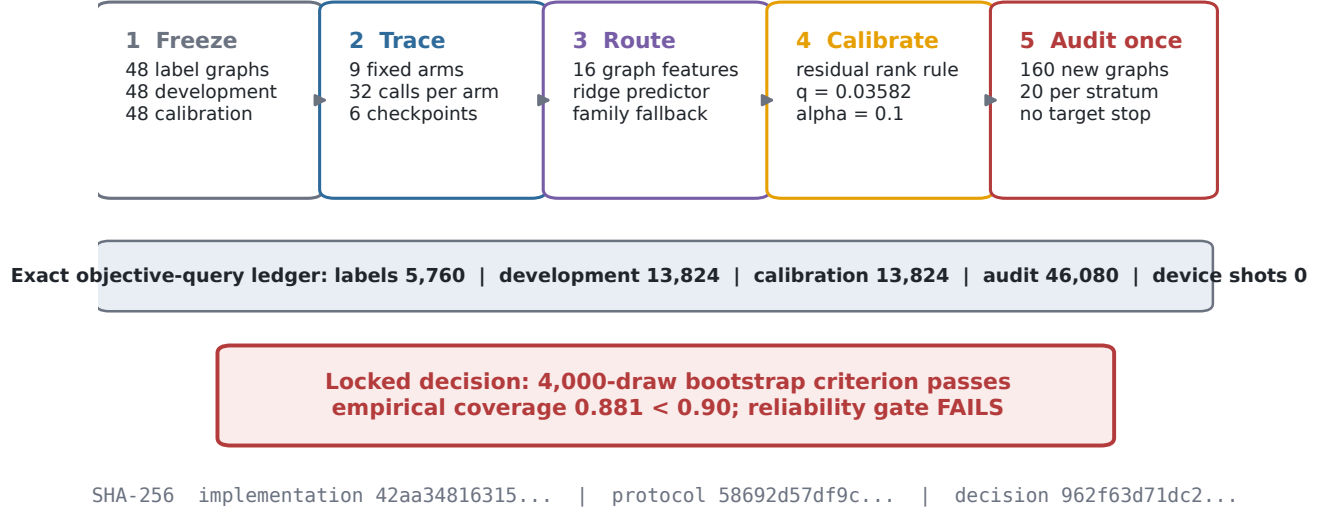


FIG. 1. **Internally locked experiment and resource ledger.** The graph list, configuration and implementation are hashed before evaluation. Development data fit both the invariant ridge selector and a family-specific fallback; calibration data set one upper residual order statistic. The decision is written before the 160-graph audit is evaluated. All arms receive the same target-free 32-query ceiling. The confirmatory reliability gate fails because descriptive empirical coverage is below its prespecified threshold.

On 64 new audit graphs its mean difference was -0.009370 with stratified 95% interval $[-0.016454, -0.002500]$. It accepted 10 graphs, harmed 1, and achieved descriptive coverage 0.9375. It passed every frozen follow-up criterion. This result was used as model-selection evidence only: it fixed the nine arms, features, ridge penalty, family fallbacks, residual-rank rule, checkpoints, bootstrap procedure and gate. No development number is treated as confirmatory.

We then created a configuration identical in all training, development and calibration seeds and all method hyperparameters, changing only the study label to confirmatory, the audit count from 8 to 20 graphs per family-size stratum, and the audit seed range. The resulting 160 graphs were generated from a new range and evaluated once (Fig. 2).

D. The confirmatory mean improves, but the reliability gate fails

On the 160 confirmatory graphs, the strongest global arm was the $d_t = 0.75$ schedule with mean AUC 0.17277. The family fallback improved this to 0.17207. The abstaining deployment achieved 0.16930; the per-instance portfolio oracle, which is unavailable at deployment, achieved 0.16048. Thus the portfolio contained opportunity, but most of it could not be selected safely.

The primary paired difference between gated deployment and family fallback was -0.002768 . A bootstrap that resampled within each family-by-size stratum 4,000 times gave percentile endpoints $[-0.005696, -0.00008178]$. Both endpoints from that prespecified run are negative, with the upper endpoint close to zero. Reasonable alternative resampling seeds and interval constructions do not uniformly retain a negative upper endpoint. We therefore report the exact locked-bootstrap output, not a claim of robust statistical significance or a conformal guarantee. The effect is not operationally broad: only 12/160 graphs (0.075) were routed away from fallback.

Four accepted routes were harmful, a joint rate of 0.0250 and a conditional rate of 0.333 among acceptances. All four harms occurred in the Barabási–Albert family, where the gated policy worsened mean AUC by $+0.003764$. Random-regular graphs supplied most of the improvement (-0.013269 ; five acceptances and no harms). Watts–Strogatz improved by -0.001567 with one acceptance, and Erdős–Rényi always used its fallback.

Most importantly, descriptive empirical coverage was 0.88125 (141 of 160), below the prespecified 0.90 requirement. The observed joint harm rate remained below α , but the mechanical decision required every criterion. It therefore returned `confirmatory_followup_ready=false` and

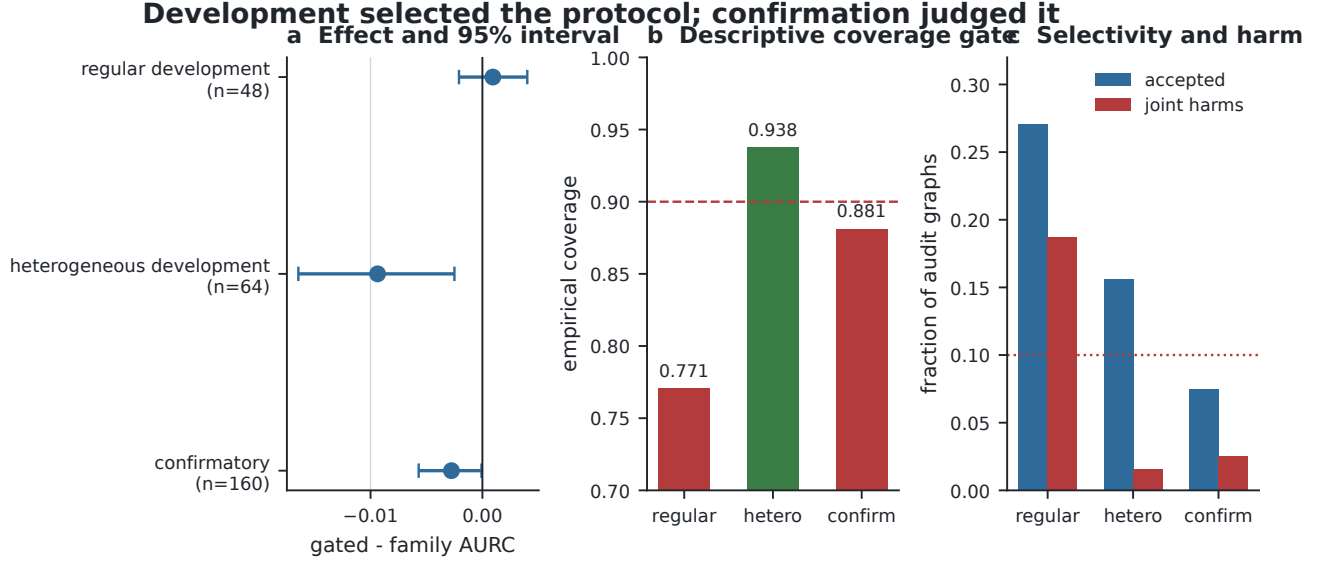


FIG. 2. **Development-to-confirmation progression.** **a**, Mean gated-minus-family-fallback AURC and stratified 95% bootstrap intervals. **b**, Descriptive empirical one-sided coverage; the dashed line is the prespecified 0.90 requirement. **c**, fractions accepted and jointly harmed. The heterogeneous development result triggered confirmation; the confirmatory coverage result failed the locked gate.

`nmi_claim_ready=false`. We do not alter that rule after observing the audit.

E. Point coverage, marginal harm and selective harm differ

The frozen gate compares a point estimate with 0.90. A point miss does not by itself show that nominal coverage is false. Under an explicitly hypothetical iid Bernoulli model with success probability 0.90, the lower-tail probability of observing at most 141 covered graphs in 160 trials is 0.249. The realized coverage is therefore statistically compatible with 0.90 under that model, even though it fails the frozen mechanical rule.

Point estimates are also weaker than high-confidence risk control. Exact one-sided 95% Clopper–Pearson upper bounds [33] distinguish three event rates. Coverage failure has 19 events among 160 graphs (rate 0.119, upper bound 0.169), so it does not certify risk at most 0.10. Under that same hypothetical iid model, joint harm has four events among all 160 graphs (0.025, upper bound 0.056), which would pass a 0.10 marginal-risk gate. Conditional harm has four events among only 12 accepted routes (0.333, upper bound 0.609), so accepted-route safety is not established. These calculations were added after the audit and are sensitivity diagnostics, not replacements for its frozen decision. Fixed quotas and sequential de-duplication also invalidate the iid premise needed to interpret them as formal confidence statements for this experiment.

F. Anytime behavior and resource accounting

Figure 3b shows mean normalized regret at all six checkpoints. The family fallback is better than the global arm early in the trajectory; the abstaining policy improves the mean primarily through a small number of random-regular instances. At 32 calls all three controls converge, while the unavailable per-checkpoint oracle remains lower. AURC rewards this early separation rather than only the terminal value.

The legacy GCTR arm was not competitive: its confirmatory mean AURC was 0.28577, compared with 0.17277 for the $d_t = 0.75$ schedule and 0.17381 for $d_t = 0.50$. Concentration, random multistart and SPSA recorded 0.29496, 0.30629 and 0.30699, respectively. These values reinforce the original negative control: angle-distribution learning and trust-region geometry did not beat simple schedules in this regime.

The confirmatory resource ledger contains 5,760 offline objective queries for 48 angle labels, 13,824 queries each for development and calibration portfolio traces, and 46,080 audit queries (nine arms times 160 graphs times 32 calls). These are experiment-construction costs. Operational deployment would execute one selected or fallback arm for at most 32 objective calls after computing cheap graph features; it would not evaluate all nine arms. The study used no shots, devices, compilation or queue time.

TABLE I. **Confirmatory results by graph family.** Fallback and gated columns are mean AURC. Δ is gated minus family fallback; negative is better. A harm is an accepted route with positive realized Δ .

Stratum	N	Fallback	Gated	Δ	Accepted	Harms
Erdős–Rényi	40	0.16439	0.16439	+0.00000	0	0
random regular	40	0.17223	0.15896	-0.01327	5	0
Barabási–Albert	40	0.18775	0.19151	+0.00376	6	4
Watts–Strogatz	40	0.16391	0.16234	-0.00157	1	0
All	160	0.17207	0.16930	-0.00277	12	4

TABLE II. **Prespecified confirmatory gate.** Every row had to pass. The locked bootstrap endpoints are negative and the joint harm rate is below α , but descriptive empirical coverage fails its requirement.

Criterion	Required	Observed	Result
Portfolio opportunity	≥ 0.05	0.071	pass
Global-baseline interval upper end	< 0	-0.000646	pass
Family-fallback interval upper end	< 0	-0.00008178	pass
Accepted audit graphs	≥ 10	12	pass
One-sided empirical coverage	≥ 0.90	0.88125	fail
Joint harm rate	≤ 0.10	0.0250	pass

II. DISCUSSION

This study separates three statements that are often merged. First, a portfolio can contain instance-level opportunity: the confirmatory oracle gap is nonzero. Second, a learned selector can lower the observed aggregate mean: the locked 4,000-resample percentile endpoints are negative, but that endpoint result is not robust to reasonable alternative inference. Third, a selector can be reliable under a prespecified deployment rule. The first is supported, the second is limited to the exact locked calculation, and the third fails.

The negative decision is informative because the protocol was designed to make failure visible. A global TQA comparison would have credited the router partly for re-discovering that graph families prefer different schedules. Replacing it with a development-selected family fallback removes much of that easy gain. Abstention then reduces exposure but does not make the residual model conditionally valid. The result also shows why a marginal harm rate and conditional safety answer different questions: four harms are only 2.5% of the full audit, but one third of accepted interventions. Exact-binomial sensitivity bounds sharpen that distinction: marginal joint harm can be small while the accepted set is too small and too failure-prone to support a selective-safety claim.

Several mechanisms may explain the family asymmetry. The 16 summaries are invariant and transparent, but they can alias graphs with different QAOA landscapes. The ridge selector is intentionally low-capacity and may underfit interactions among density, clustering and spectrum. Conversely, replacing it after seeing the audit would create selection bias. The calibration split has only 48 graphs, so its upper residual order statistic is coarse. Moreover, the fixed family-by-size quotas and

sequential cross-split non-isomorphism filter mean that calibration and audit residuals are not literally exchangeable under the standard pooled theorem. Even under idealized exchangeability, split conformal would be marginal and would not guarantee coverage within the Barabási–Albert subgroup or conditional on acceptance [24, 27]. The BA degradation is therefore a failure mode to explain in a future, newly frozen study, not a cue for post hoc threshold adjustment here.

The work is also bounded by its quantum evidence. All results are exact statevector simulations of unweighted MaxCut at $p = 2$ and $n \in \{12, 14\}$. The exact optimum is used to normalize regret. Shot noise can reorder arms, SPSA is designed for stochastic rather than exact objectives [15], and device compilation and latency can dominate an evaluation count. Recent work demonstrates parameter transfer and online learning on much larger or hardware-native problems [17, 18]; the present experiment neither matches that scale nor supplies hardware evidence. No quantum advantage, large-scale classical advantage or end-to-end wall-clock gain is claimed.

Offline costs are unequal. The concentration, k-nearest-neighbor and legacy GCTR arms share 48 labels produced by 5,760 exact objective calls. The linear schedules, random multistart and SPSA require none. The audit pays to evaluate every arm only because ground truth is needed to judge routing; a deployed router would not. Any claim of amortized advantage must specify how many future instances repay the labels, model fit and portfolio audit. Label-free graph-conditioned trajectory learning is an important alternative [12].

A credible next stage would be a new protocol, not an update to this audit. It would increase calibration and audit sizes, include unknown-family or family-agnostic fallbacks, and use a sampling design and group-conditional method with a justified coverage theorem. It would also

One-shot confirmatory audit: a small average gain without reliable routing

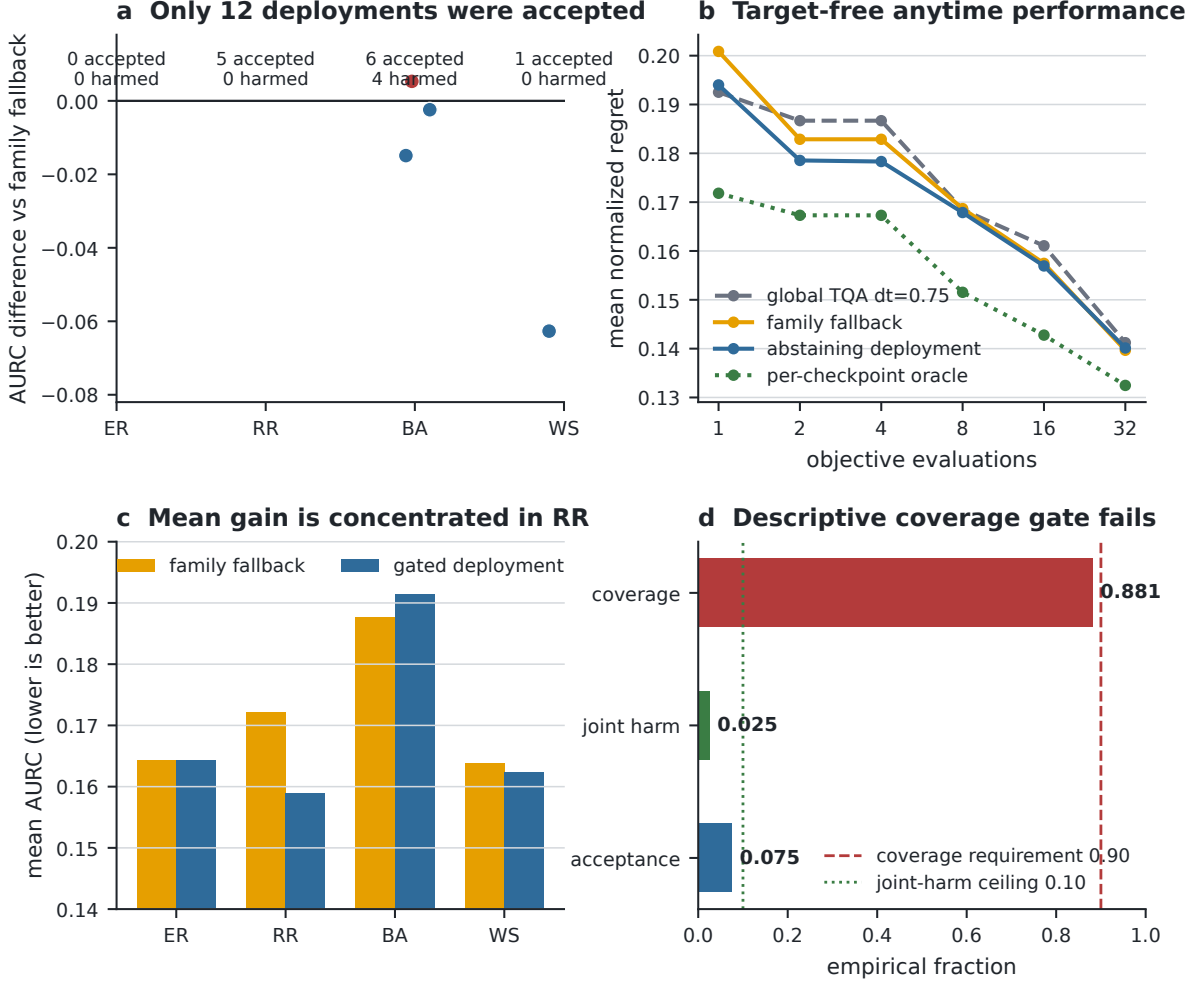


FIG. 3. **Confirmatory audit.** **a**, Realized AURC differences for the 12 accepted routes; red points are harmful. **b**, mean anytime regret for the global schedule, family fallback, gated deployment and unavailable per-checkpoint oracle. **c**, family means expose concentration of gains in random-regular graphs and degradation in Barabási–Albert graphs. **d**, the descriptive empirical coverage criterion fails even though the joint harm rate is below its numerical ceiling.

evaluate noisy and repeated-shot objectives and report total shots, latency and classical preprocessing. A method change must invalidate the current decision hash and use new audit seeds. Until such evidence exists, the strongest conclusion is methodological: a prespecified reliability audit can stop a small positive mean from being overstated as reliable learned optimization.

III. METHODS

A. Graph distributions and frozen splits

The heterogeneous protocol uses four unweighted random-graph families at $n = 12$ and $n = 14$.

Erdős–Rényi edge probability is sampled uniformly from $[0.15, 0.75]$. Random-regular degree is sampled from the valid members of $\{2, 3, 4, 5, 6\}$. Barabási–Albert attachment is sampled from $\{1, 2, 3, 4\}$. Watts–Strogatz degree is sampled from $\{2, 4, 6, 8\}$ and rewiring probability uniformly from $[0.02, 0.95]$ [28–30].

Each family-size stratum contains six angle-training, six development, six calibration and 20 confirmatory audit graphs: 48, 48, 48 and 160 graphs, respectively. A candidate is rejected if its labeled edge hash duplicates a prior graph. A Weisfeiler–Lehman-plus-degree fingerprint screens possible isomorphs, which are then checked by exact NetworkX isomorphism [31]. Generator parameters, seeds, edge hashes, exact MaxCut values and invariant fingerprints are stored in the frozen protocol.

The successful heterogeneous development config-

uration and the confirmatory configuration share angle-training seed start 1,310,000; development start 1,410,000; calibration start 1,510,000; model/optimizer seed 20,260,719; and all hyperparameters. Development used audit seeds from 1,610,000. Confirmation used a new range from 1,810,000 and increased the audit count from eight to 20 per stratum. Protocol construction rejects isomorphic overlap across every split.

B. Angle labels and legacy model

Each angle-training graph receives three random-start coordinate searches, each capped at 40 exact calls. The best four-angle vector is the label, costing 120 queries per graph and 5,760 in total. The coordinate-wise median label is the concentration arm. The nearest-neighbor arm standardizes the 16 invariant features using angle-training statistics and transfers the label with smallest Euclidean feature distance.

The legacy arm retains the earlier graph-conditioned idea as a deliberately strong learned comparator. A three-layer graph isomorphism network with width 48 and six Laplacian spectral node features predicts diagonal-Gaussian mean and log variance [19]. It is trained for 80 Adam epochs at learning rate 0.005 with heteroscedastic Gaussian negative log likelihood and a small log-variance penalty. Its mean starts coordinate search, the concentration vector is an additional counted seed, predicted standard deviations scale steps, and a radius-two Mahalanobis ball limits proposals. The region recenters on the best seed. No development, calibration or audit result is used to fit this model.

C. Optimizer arms and exact accounting

The four schedule arms use midpoint grid $g_j = (j - 1/2)/p$ and angles $\gamma_j = g_j d_t$, $\beta_j = (1 - g_j) d_t$. Concentration, schedule, k-nearest neighbor and legacy arms use the same best-so-far coordinate pattern search: initial step 0.3; both signed proposals for each coordinate; halve the step after a non-improving sweep; and, once the step falls below 10^{-3} , restart near the incumbent so that the target-free trace consumes all 32 calls. Random multistart divides the budget across four random QAOA starts. SPSA maximizes the objective with $a = 0.18$, $c = 0.12$, decay exponents 0.602 and 0.101, and records both signed evaluations [15].

The `BudgetedObjective` raises an exception on a 33rd call. A trace is valid only if it contains exactly 32 calls and monotonically non-decreasing best-so-far values. It stores checkpoint parameters and values, `objective_queries=32`, `shots_used=0` and back-end `exact_statevector`. Unit tests cover budget exhaustion, monotonicity, SPSA and multistart accounting.

D. Selector, residual-order-statistic abstention and gate

The selector is fit once to the 48 development rows and nine arm utilities. Every feature is standardized with the development mean and standard deviation; constant scales are set to one. Ridge regression includes an unpenalized intercept and penalty one. Family fallbacks are selected by development mean as in Eq. (6).

The selected arm for each of 48 calibration graphs is determined without its utility. Residuals in Eq. (7) then set the one-sided order statistic. The margin is zero. A standard split-conformal statement would assume exchangeable calibration and future residuals with a decision function fixed independently of both outcome sets. The realized protocol does not meet exact pooled exchangeability: it fixes the count in every family-by-size stratum and accepts graphs sequentially only after excluding isomorphs from all earlier splits. We therefore invoke no finite-sample coverage theorem. The order statistic is a prespecified abstention heuristic, and its audit coverage is a descriptive gate rather than a calibrated risk guarantee. Even the idealized pooled statement would say nothing about exact subgroup or acceptance-conditional coverage. Modern selection-conditional conformal procedures instead construct calibration sets around the selection event and control false coverage statements in online settings [25]. The pooled residual heuristic used here is not such a procedure, so its marginal motivation must not be read as accepted-route safety.

The confirmatory follow-up flag required all of the following: portfolio opportunity at least 5% of global-baseline AURC; upper end of the stratified 95% interval below zero versus both global and family baselines; at least ten accepted graphs; empirical one-sided coverage at least 0.90; and joint harm rate at most 0.10. `nmi_claim_ready` additionally required a confirmatory study label. The code evaluates these Boolean criteria without manual interpretation. That legacy field name is only a mechanical experiment label; it is not, and cannot be, a determination of journal readiness.

E. Statistical analysis

The experimental unit is a graph. The primary estimand is the arithmetic mean of paired per-graph AURC differences between the deployed arm and the frozen family fallback. The 95% interval is a nonparametric bootstrap with 4,000 resamples, sampling with replacement separately inside each of the eight family-by-size strata and then taking the 2.5th and 97.5th percentiles [32]. The bootstrap seed is fixed in the configuration. No post hoc p -value or multiplicity-adjusted family inference is reported. Per-family means diagnose heterogeneity and are descriptive.

Descriptive empirical coverage is the fraction satisfying $u_s - u_b \leq U(x)$, whether or not the route is ac-

TABLE III. **Confirmatory split and exact-query ledger.** Per family-size is the number in each of eight family-by-size strata. Trace queries include all nine arms at 32 calls per graph.

Split	Per family-size	Graphs	Exact queries	Role
Angle labels	6	48	5,760	legacy-arm training
Development	6	48	13,824	routing and fallback
Calibration	6	48	13,824	residual order statistic
Confirmatory audit	20	160	46,080	one-shot decision

cepted. A joint harm is **accepted** and deployed-minus-fallback AURC greater than zero. The conditional harm rate divides harms by accepted routes and is descriptive; it was not a gate criterion. Opportunity is $(\bar{u}_{\text{global}} - \bar{u}_{\text{oracle}})/\bar{u}_{\text{global}}$.

The retrospective risk sensitivity uses the one-sided exact Clopper–Pearson upper bound

$$p_U = \text{Beta}_{1-\eta}^{-1}(k+1, n-k) \quad (9)$$

for $k < n$ events among n units at confidence $1 - \eta = 0.95$; $p_U = 1$ when $k = n$. The nominal-coverage tail probability is $\Pr\{\text{Binomial}(160, 0.90) \leq 141\}$. These calculations assume independent identically distributed Bernoulli events. They are deliberately labelled sensitivity diagnostics because the locked sampling design does not establish that premise.

F. Internal locking, chronology and provenance

Canonical JSON uses sorted keys and compact separators before SHA-256 hashing. The confirmatory configuration hash begins `a0e66aa4c41f`; protocol hash `58692d57df9c`; implementation hash `42aa34816315`; and frozen decision hash `962f63d71dc2`. The decision stores the selector coefficients, family arms, residual order statistic, non-audit trace hash and legacy model hash. It is written after development/calibration and before audit evaluation. The runner refuses to proceed if `traces.jsonl` already exists, requiring a new directory and protocol for any repeat.

The manifest hashes angle labels, model checkpoint, non-audit traces, decision, all traces and summary. Validation recomputes file, protocol, decision, configuration and implementation hashes. The manuscript plotting script first validates all three studies, reads only locked JSON/JSONL, and records hashes of its generated PDFs, PNGs, tables and numerical macros. Running it twice yields the same artifact hashes in the recorded environment.

These hashes provide tamper evidence and the runner enforces an internal execution order. They are not an independent timestamped preregistration: the protocol, frozen decision and audit outcomes first entered the public Git history together in one repository commit. Repository history therefore cannot independently verify that the decision existed before the outcomes. A future confirmatory study must publish an immutable external

timestamp for the executable protocol, trained selector, calibration thresholds and analysis code before any audit outcome is generated.

G. Use of generative artificial intelligence

A generative language model assisted with software implementation, numerical analysis, figure and table generation, and manuscript drafting. Molena Huynh is the sole author, verified the code, locked records, results, citations and scientific claims, and retains full responsibility for the final work.

IV. DATA AVAILABILITY

All graph identities, generator parameters, objective traces, exact query counts, split summaries and hashes used here are included in `portfolio_results/`. The controlling confirmation is `portfolio_results/heterogeneous_confirmatory_v1/`. No external or proprietary dataset is used. Exact MaxCut values and expectation ratios are small-instance simulation data, not measurements from a quantum processor.

V. CODE AVAILABILITY

The installable package is in `specops_gctr/`. The portfolio entry point is `gctr-portfolio`; configurations are in `configs/`. The deterministic manuscript generator is `manuscript/generate_portfolio_artifacts.py`. A reviewer can verify the locked audit and regenerate display artifacts with

```
python run.py portfolio \
--config configs/portfolio_heterogeneous_confirmatory.
  json \
--output-dir portfolio_results/
  heterogeneous_confirmatory_v1 \
--validate-only
python manuscript/generate_portfolio_artifacts.py
gctr-risk-audit \
  portfolio_results/heterogeneous_confirmatory_v1/summary.
  json
pytest -q
```

The existing audit cannot be overwritten by the runner. The exact-binomial diagnostic is implemented separately in `specops_gctr/risk_control.py`; it does not

alter the frozen result. The file `configs/prospective_risk_control_v2.design.json` is an unexecuted design specification, explicitly marked unregistered and incomplete. It supports planning only and contains no future outcome.

AUTHOR CONTRIBUTIONS

M.H. conceived the study, designed the methods, implemented and verified the software, conducted the analysis, interpreted the results, and wrote the manuscript.

COMPETING INTERESTS

The author declares no competing interests.

REFERENCES

-
- [1] Farhi, E., Goldstone, J. & Gutmann, S. A quantum approximate optimization algorithm. *arXiv* <https://arxiv.org/abs/1411.4028> (2014).
 - [2] Hadfield, S. *et al.* From the quantum approximate optimization algorithm to a quantum alternating operator ansatz. *Algorithms* **12**, 34 (2019).
 - [3] Zhou, L., Wang, S.-T., Choi, S., Pichler, H. & Lukin, M. D. Quantum approximate optimization algorithm: performance, mechanism, and implementation on near-term devices. *Phys. Rev. X* **10**, 021067 (2020).
 - [4] Cerezo, M. *et al.* Variational quantum algorithms. *Nat. Rev. Phys.* **3**, 625–644 (2021).
 - [5] Bharti, K. *et al.* Noisy intermediate-scale quantum algorithms. *Rev. Mod. Phys.* **94**, 015004 (2022).
 - [6] Brandao, F. G. S. L., Broughton, M., Farhi, E., Gutmann, S. & Neven, H. For fixed control parameters the quantum approximate optimization algorithm’s objective function value concentrates for typical instances. *arXiv* <https://arxiv.org/abs/1812.04170> (2018).
 - [7] Akshay, V., Rabinovich, D., Campos, E. & Biamonte, J. Parameter concentrations in quantum approximate optimization. *Phys. Rev. A* **104**, L010401 (2021).
 - [8] Egger, D. J., Marecek, J. & Woerner, S. Warm-starting quantum optimization. *Quantum* **5**, 479 (2021).
 - [9] Jain, N., Coyle, B., Kashefi, E. & Kumar, N. Graph neural network initialisation of quantum approximate optimisation. *Quantum* **6**, 861 (2022).
 - [10] Jiang, L., Zhang, C. & Chen, F. QSeer: a quantum-inspired graph neural network for parameter initialization in quantum approximate optimization algorithm circuits. *arXiv* <https://arxiv.org/abs/2505.06810> (2025).
 - [11] Khairy, S., Shaydulin, R., Cincio, L., Alexeev, Y. & Balaprakash, P. Learning to optimize variational quantum circuits to solve combinatorial problems. In *Proc. AAAI Conf. Artif. Intell.* **34**, 2367–2375 (2020).
 - [12] Nguyen, K. X. & Safro, I. Graph-conditioned meta-optimizer for QAOA parameter generation on multiple problem classes. *arXiv* <https://arxiv.org/abs/2604.25275> (2026).
 - [13] Huynh, M. Query-efficient quantum approximate optimization via graph-conditioned trust regions. *arXiv:2604.24803v1* <https://arxiv.org/abs/2604.24803v1> (2026).
 - [14] Bonet-Monroig, X. *et al.* Performance comparison of optimization methods on variational quantum algorithms. *Phys. Rev. A* **107**, 032407 (2023).
 - [15] Spall, J. C. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Trans. Autom. Control* **37**, 332–341 (1992).
 - [16] Montañez-Barrera, J. A. & Michielsen, K. Toward a linear-ramp QAOA protocol: evidence of a scaling advantage in solving some combinatorial optimization problems. *npj Quantum Inf.* **11**, 131 (2025).
 - [17] Shaydulin, R., Lotshaw, P. C., Larson, J., Ostrowski, J. & Humble, T. S. Parameter transfer for quantum approximate optimization of weighted MaxCut. *ACM Trans. Quantum Comput.* **4**, 19 (2023).
 - [18] O’Leary, T. *et al.* Efficient online quantum circuit learning with no upfront training. *Commun. Phys.* **8**, 514 (2025).
 - [19] Xu, K., Hu, W., Leskovec, J. & Jegelka, S. How powerful are graph neural networks? In *International Conference on Learning Representations* (2019).
 - [20] Rice, J. R. The algorithm selection problem. *Adv. Comput.* **15**, 65–118 (1976).
 - [21] Cawley, G. C. & Talbot, N. L. C. On over-fitting in model selection and subsequent selection bias in performance evaluation. *J. Mach. Learn. Res.* **11**, 2079–2107 (2010).
 - [22] Moussa, C., Calandra, H. & Dunjko, V. To quantum or not to quantum: towards algorithm selection in near-term quantum optimization. *Quantum Sci. Technol.* **5**, 044009 (2020).
 - [23] Balcan, M.-F., Sandholm, T. & Vitercik, E. Generalization in portfolio-based algorithm selection. In *Proc. AAAI Conf. Artif. Intell.* **35**, 12225–12232 (2021).
 - [24] Angelopoulos, A. N. & Bates, S. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv* <https://arxiv.org/abs/2107.07511> (2021).
 - [25] Bao, Y., Huo, Y., Ren, H. & Zou, C. CAP: a general algorithm for online selective conformal prediction with FCR control. *J. Mach. Learn. Res.* **26**, 1–74 (2025).
 - [26] Romano, Y., Patterson, E. & Candès, E. J. Conformalized quantile regression. In *Advances in Neural Information Processing Systems* **32** (2019).

- [27] Barber, R. F., Candès, E. J., Ramdas, A. & Tibshirani, R. J. Conformal prediction beyond exchangeability. *arXiv* <https://arxiv.org/abs/2202.13415> (2022).
- [28] Erdős, P. & Rényi, A. On random graphs. *Publ. Math. Debrecen* **6**, 290–297 (1959).
- [29] Barabási, A.-L. & Albert, R. Emergence of scaling in random networks. *Science* **286**, 509–512 (1999).
- [30] Watts, D. J. & Strogatz, S. H. Collective dynamics of small-world networks. *Nature* **393**, 440–442 (1998).
- [31] Hagberg, A. A., Schult, D. A. & Swart, P. J. Exploring network structure, dynamics, and function using NetworkX. In *Proc. 7th Python in Science Conference*, 11–15 (2008).
- [32] Efron, B. & Tibshirani, R. J. *An Introduction to the Bootstrap* (Chapman & Hall/CRC, 1993).
- [33] Clopper, C. J. & Pearson, E. S. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika* **26**, 404–413 (1934).